

Project Design Phase – II

Solution Requirements (Functional & Non-functional)

Date	
Team ID	
Project Name	Credit Card Approval Prediction
Maximum marks	

Functional Requirements

Functional requirements describe what the system should do.

Requirement ID	Functional Requirement
FR1	The system should allow users to enter applicant details.
FR2	The system should validate the entered information.
FR3	The system should preprocess the input data before prediction.
FR4	The system should load the trained machine learning model.
FR5	The system should predict whether the credit card application is Approved or Rejected.
FR6	The system should display the prediction result to the user.
FR7	The system should allow users to make multiple predictions.

Non-Functional Requirements

Non-functional requirements describe how the system should perform.

Requirement ID	Non-Functional Requirement
NFR1	The system should provide prediction results within a few seconds.
NFR2	The application should have a simple and user-friendly interface.
NFR3	The system should provide accurate prediction results.
NFR4	The application should be reliable and stable.
NFR5	The system should be easy to maintain and update.
NFR6	The application should protect user data during processing.

Project Design Phase-II

Data Flow Diagram & User Stories

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Data Flow (Simplified Process)

This sequence outlines how data moves from the user through the prediction system:

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- User Authentication:** User logs into the portal to access the application form.
- Data Submission:** User enters financial details (income, credit score, debt, etc.) and uploads necessary documents.
- Data Preprocessing:** The system cleans the input data and handles missing values for the machine learning model.
- Prediction Engine:** The preprocessed data is passed to the trained Machine Learning model to evaluate approval probability.
- Result Visualization:** The approval status (Approved/Rejected) and risk score are displayed to the user on the dashboard.

User Stories

Following the provided template, here are the user stories for the credit card approval system:

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User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Applicant	Registration	USN-1	As a user, I can register using my email and personal details.	I can access my applicant dashboard.	High	Sprint-1
Applicant	Application Submission	USN-2	As a user, I can fill out and submit my financial details for credit card approval.	System confirms successful submission.	High	Sprint-1
System / ML Model	Prediction	USN-3	As a system, I want to process applicant data through a model to predict approval status.	Prediction is generated within seconds of submission.	High	Sprint-2
Applicant	Dashboard	USN-4	As a user, I want to view the status of my application (Pending/Approved/Rejected).	Status is clearly visible on the main dashboard.	Medium	Sprint-2
Admin / Bank Staff	Data Management	USN-5	As an admin, I can view all submitted applications and the model's risk assessment.	Access to a list of all applicants and their scores.	Medium	Sprint-3
Applicant	Notification	USN-6	As a user, I want to receive an email notification once a decision is made.	Email is sent automatically upon status change.	Low	Sprint-3

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